

Reaper | HackTheBox Writeup

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Reaper
Very Easy

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★ 4.5 (114)
Sherlock Rating

🏆 225
XP reward

Sherlock Scenario

Our SIEM alerted us to a suspicious logon event which needs to be looked at immediately . The alert details were that the IP Address and the Source Workstation name were a mismatch .You are provided a network capture and event logs from the surrounding time around the incident timeframe. Corelate the given evidence and report back to your SOC Manager.

Process:

Task 1

Hint

What is the IP Address for Forela-Wkstn001?

172.17.79.129

Submit Task

Task 2

Hint

What is the IP Address for Forela-Wkstn002?

172.17.79.136

Submit Task

While analyzing the PCAP, I noticed NBNS traffic. NBNS (NetBIOS Name Service) is a protocol used to resolve NetBIOS hostnames to IP addresses on a local network. Because NBNS responses can associate a hostname with a specific IP address, this traffic can help confirm which hostname belongs to a particular IP address observed in the investigation.

Task 3
Hint

What is the username of the account whose hash was stolen by attacker?

arthur.kyle

Submit Task

Task 4
Hint

What is the IP Address of Unknown Device used by the attacker to intercept credentials?

172.17.79.135

Submit Task

Instead of a direct, legitimate authentication flow between the user and the server, the traffic was hijacked by an unauthorized middleman IP address (`172.17.79.135`). This began when the victim machine broadcasted an **NBNS query** looking for a resource; the unknown IP aggressively responded to the request despite having **no legitimate hostname or DNS association** to that name.

No.	Time	Source	Destination	Protocol	Length	Info
580	4.471779579	172.17.79.1	172.17.79.135	NBNS	199	Name query response NBSTAT
3	0.001463822	172.17.79.129	172.17.79.2	NBNS	110	Refresh NB FORELA-WKSTN001<20>
28	1.499287702	172.17.79.129	172.17.79.2	NBNS	110	Refresh NB FORELA-WKSTN001<20>
294	3.014968616	172.17.79.129	172.17.79.2	NBNS	110	Refresh NB FORELA-WKSTN001<20>
586	4.473304535	172.17.79.129	172.17.79.135	NBNS	199	Name query response NBSTAT
612	4.515129182	172.17.79.129	172.17.79.2	NBNS	110	Refresh NB FORELA-WKSTN001<00>
622	6.062738111	172.17.79.129	172.17.79.2	NBNS	110	Refresh NB FORELA-WKSTN001<00>
623	7.577220003	172.17.79.129	172.17.79.2	NBNS	110	Refresh NB FORELA-WKSTN001<00>
631	9.077505400	172.17.79.129	172.17.79.2	NBNS	110	Refresh NB FORELA<00>
632	10.592792735	172.17.79.129	172.17.79.2	NBNS	110	Refresh NB FORELA<00>
636	12.108452017	172.17.79.129	172.17.79.2	NBNS	110	Refresh NB FORELA<00>

Deceived by this spoofed response, the victim machine route its traffic to the unknown IP, unknowingly sending its initial NTLM negotiation request there. The unassociated IP then immediately mirrored that request by establishing a separate, parallel connection to the actual target server (`172.17.79.129`). By acting as a transparent proxy, the unknown IP relayed the server's cryptographic challenges back to the victim and subsequently forwarded the victim's valid authentication response (`FORELA\arthur.kyle`) back to the server, successfully logging in as that user.

No.	Time	Source	Destination	Protocol	Length	Info
1408	112.555962896	172.17.79.136	172.17.79.4	SMB2	3452	Session Setup Request
1386	112.552805169	172.17.79.136	172.17.79.4	SMB2	274	Negotiate Protocol Request
1255	98.048736924	172.17.79.136	172.17.79.135	SMB2	594	Session Setup Request, NTLMSSP_AUTH, User: FORELA\arthur.kyle
1241	97.997053284	172.17.79.136	172.17.79.135	SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE
1239	97.996276172	172.17.79.136	172.17.79.135	SMB2	126	Session Logoff Request
1235	97.995020118	172.17.79.136	172.17.79.135	SMB2	212	Ioctl Request FSCTL_VALIDATE_NEGOTIATE_INFO
1231	97.993065158	172.17.79.136	172.17.79.135	SMB2	146	Tree Connect Request, Tree: '\\D\IPC\$'
1213	97.987030692	172.17.79.136	172.17.79.135	SMB2	664	Session Setup Request, NTLMSSP_AUTH, User: FORELA\arthur.kyle
1209	97.984658985	172.17.79.136	172.17.79.135	SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE
1199	97.977977002	172.17.79.136	172.17.79.135	SMB2	146	Tree Connect Request, Tree: '\\D\IPC\$'
1197	97.976683617	172.17.79.136	172.17.79.135	SMB2	635	Session Setup Request, NTLMSSP_AUTH, User: FORELA\arthur.kyle

No.	Time	Source	Destination	Protocol	Length	Info
1184	97.971106716	172.17.79.135	172.17.79.136	TCP	54	445 → 50145 [ACK] Seq=1 Ack=74 Win=32128 Len=0
1185	97.972371959	172.17.79.135	172.17.79.136	SMB2	228	Negotiate Protocol Response
1196	97.976170490	172.17.79.135	172.17.79.136	SMB2	347	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP_CHALLENGE
1198	97.97669387	172.17.79.135	172.17.79.136	SMB2	139	Session Setup Response
1200	97.978446585	172.17.79.135	172.17.79.129	TCP	74	40252 → 445 [SYN] Seq=0 Win=32120 Len=0 MSS=1460 SACK_PERM TSval=1164903174 TSecr=0 WS=128
1202	97.978761025	172.17.79.135	172.17.79.129	TCP	54	40252 → 445 [ACK] Seq=1 Ack=1 Win=32128 Len=0
1203	97.978932875	172.17.79.135	172.17.79.129	SMB	127	Negotiate Protocol Request
1205	97.979694909	172.17.79.135	172.17.79.129	TCP	54	40252 → 445 [ACK] Seq=74 Ack=253 Win=32000 Len=0
1206	97.982824701	172.17.79.135	172.17.79.129	SMB2	164	Negotiate Protocol Request
1208	97.983859432	172.17.79.135	172.17.79.136	SMB2	138	Tree Connect Response, Error: STATUS_NETWORK_SESSION_EXPIRED, Tree: '\\D\IPC\$'
1210	97.985540721	172.17.79.135	172.17.79.129	SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE
1212	97.986639011	172.17.79.135	172.17.79.136	SMB2	380	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP_CHALLENGE
1214	97.987948636	172.17.79.135	172.17.79.129	SMB2	664	Session Setup Request, NTLMSSP_AUTH, User: FORELA\arthur.kyle
1230	97.992801806	172.17.79.135	172.17.79.136	SMB2	138	Session Setup Response
1232	97.993990427	172.17.79.135	172.17.79.129	SMB2	170	Tree Connect Request, Tree: '\\172.17.79.129\IPC\$'
1234	97.994795829	172.17.79.135	172.17.79.136	SMB2	138	Tree Connect Response, Tree: '\\D\IPC\$'
1236	97.995357239	172.17.79.135	172.17.79.129	SMB2	190	Create Request, File: svcctl
1238	97.995999966	172.17.79.135	172.17.79.136	SMB2	194	Ioctl Response FSCTL_VALIDATE_NEGOTIATE_INFO

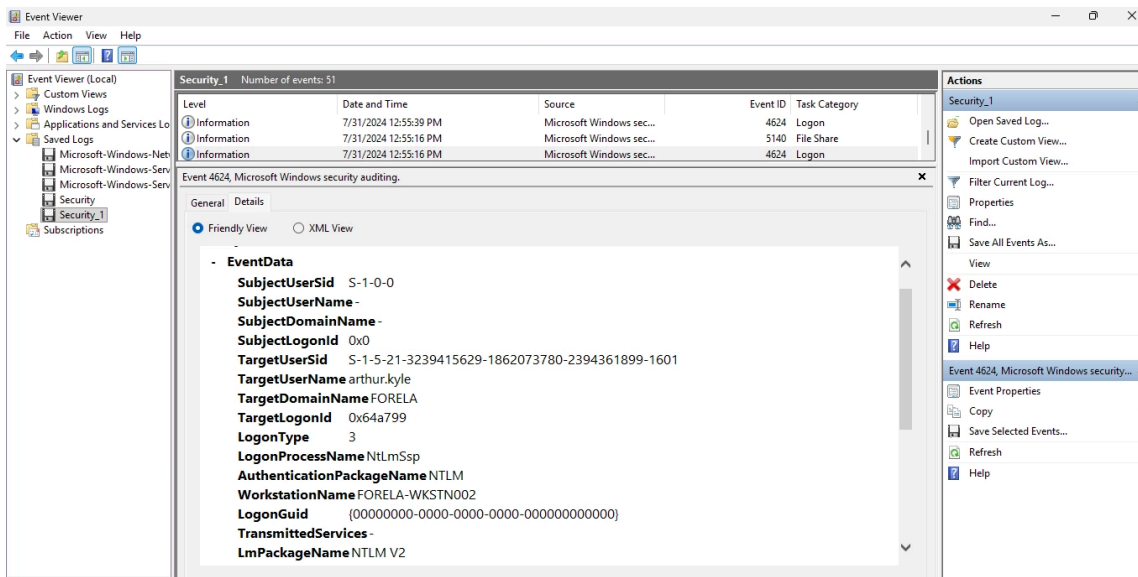
Task 5
Hint

What was the fileshare navigated by the victim user account?

Prior to the relay, the user was actively accessing network resources, specifically connecting to the share `\\DC01\Trip`.

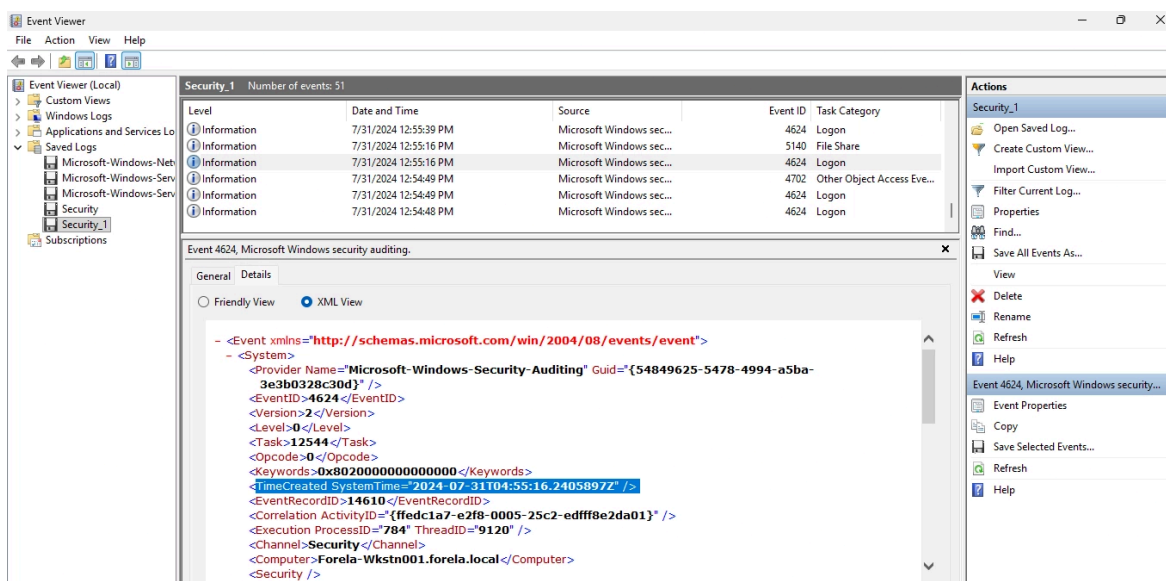
No.	Time	Source	Destination	Protocol	Length	Info
1434	112.568877903	172.17.79.136	172.17.79.4	SMB2	190	Create Request, File: srvsvc
1432	112.568175286	172.17.79.136	172.17.79.4	SMB2	152	Tree Connect Request, Tree: '\\DC01\Trip'
1430	112.567878203	172.17.79.136	172.17.79.4	SMB2	152	Tree Connect Request, Tree: '\\DC01\Trip'
1428	112.567600707	172.17.79.136	172.17.79.4	SMB2	152	Tree Connect Request, Tree: '\\DC01\Trip'
1426	112.567344810	172.17.79.136	172.17.79.4	SMB2	152	Tree Connect Request, Tree: '\\DC01\Trip'
1424	112.566899230	172.17.79.136	172.17.79.4	SMB2	152	Tree Connect Request, Tree: '\\DC01\Trip'
1422	112.566608349	172.17.79.136	172.17.79.4	SMB2	152	Tree Connect Request, Tree: '\\DC01\Trip'
1420	112.566315545	172.17.79.136	172.17.79.4	SMB2	152	Tree Connect Request, Tree: '\\DC01\Trip'
1418	112.565901644	172.17.79.136	172.17.79.4	SMB2	152	Tree Connect Request, Tree: '\\DC01\Trip'
1414	112.557146449	172.17.79.136	172.17.79.4	SMB2	202	Ioctl Request FSCTL_DFS_GET_REFERRALS, Path: \DC01\Trip
1413	112.557050971	172.17.79.136	172.17.79.4	SMB2	178	Ioctl Request FSCTL_QUERY_NETWORK_INTERFACE_INFO
1411	112.556745492	172.17.79.136	172.17.79.4	SMB2	152	Tree Connect Request, Tree: '\\DC01\IPC\$'
1408	112.555962896	172.17.79.136	172.17.79.4	SMB2	3452	Session Setup Request
1386	112.552805169	172.17.79.136	172.17.79.4	SMB2	274	Negotiate Protocol Request
1255	98.048736924	172.17.79.136	172.17.79.135	SMB2	594	Session Setup Request, NTLMSSP_AUTH, User: FORELA\arthur.kyle
1241	97.997053284	172.17.79.136	172.17.79.135	SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE

Following the interception, analysis of the Windows Security event logs (`.evtx`) confirmed the successful execution of the attack. Security Event ID 4624 recorded a successful network authentication (Logon Type 3) at `2024-07-31T04:55:16.2405897Z`.



"New Logon" more information about the log :

Account Name **arthur.kyle** , the Account Domain **FORELA** , and the Logon ID **0x64A799** .



While the target account name was **FORELA\arthur.kyle** and the workstation name was listed as **FORELA-WKSTN002** ,

Task 8
Hint

The detection was based on the mismatch of hostname and the assigned IP Address.What is the workstation name and the source IP Address from which the malicious logon occur?

FORELA-WKSTN002, 172.17.79.135
Submit Task

Task 9
Hint

At what UTC time did the the malicious logon happen?

2024-07-31 04:55:16
Submit Task

quick look up in chainsaw \ more efficient alternative to event viewer :

```

Windows PowerShell
PS C:\Users\alexa\Desktop\chainsaw_x86_64-pc-windows-msvc\chainsaw> .\chainsaw.exe search "C:\Users\alexa\Documents\Bridge\Labs\Reaper\Reaper\Security.evtx" --tau "Event.System.EventID: =4624" --tau "Event.EventData.LogonType: =3"

CHAINSAW
By WithSecure Countercept (@FranticTyping, @AlexKornitzer)

[+] Loading forensic artefacts from: C:\Users\alexa\Documents\Bridge\Labs\Reaper\Reaper\Security.evtx
[+] Loaded 1 forensic files (1.1 MiB)
[+] Searching forensic artefacts...

Event_attributes:
  xmlns: http://schemas.microsoft.com/win/2004/08/events/event
Event:
  System:
    Provider_attributes:
      Name: Microsoft-Windows-Security-Auditing
      Guid: 54849625-5478-4994-A58A-3E3B0328C38D
    EventID: 4624
    Version: 2
    Level: 0
    Task: 12544
    Opcode: 0
    Keywords: '0x8020000000000000'
    TimeCreated_attributes:
      SystemTime: 2024-07-31T04:54:48.626961Z
  
```

Task 6
Hint

What is the source port used to logon to target workstation using the compromised account?

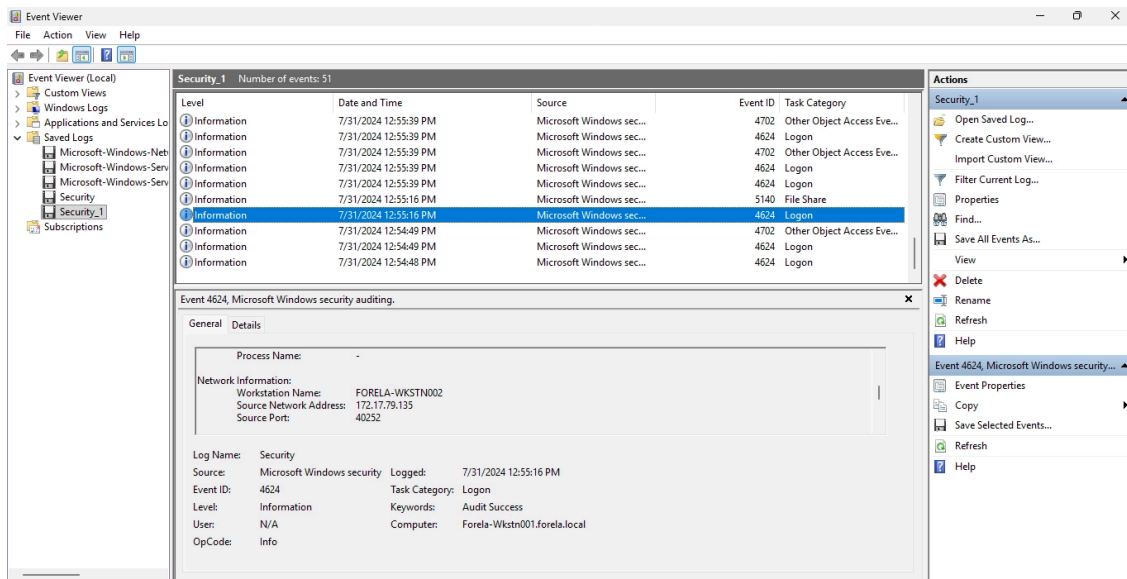
40252
Submit Task

Task 7
Hint

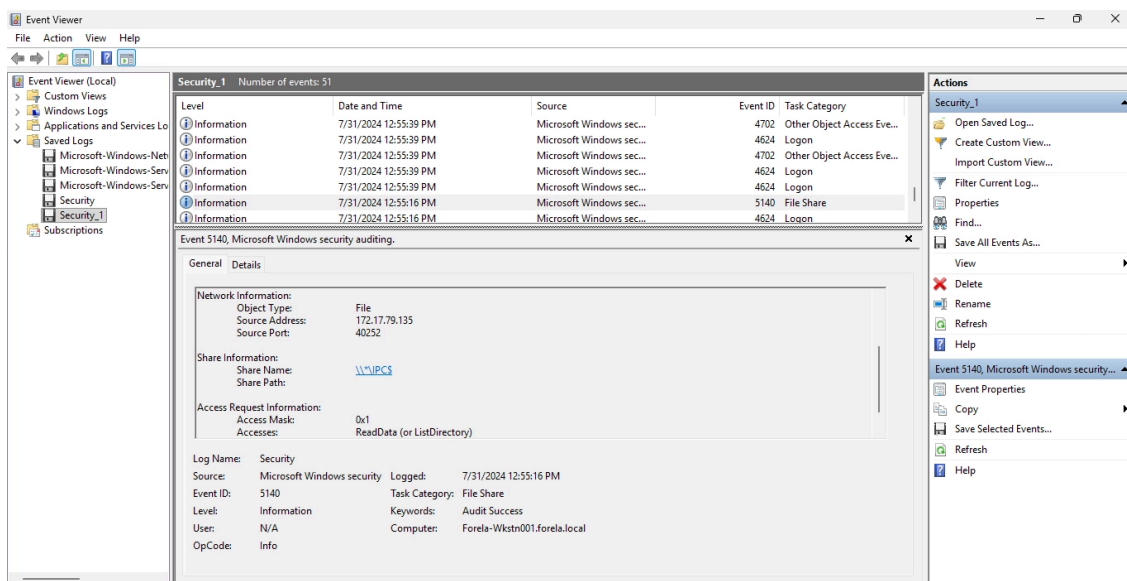
What is the Logon ID for the malicious session?

0x64A799
Submit Task

the log explicitly captured the attacker's machine as the true origin, noting a Source Network Address of **172.17.79.135** and Source Port **40252**.



Immediately following this logon, Security Event ID 5140 recorded that the attacker utilized these relayed credentials to access a network share object, specifically targeting the IPC share named ***\IPC\$** to establish a named pipe connection for further reconnaissance and lateral movement.



Task 10Hint

What is the share Name accessed as part of the authentication process by the malicious tool used by the attacker?

***\IPC\$**Submit Task

reference:

<https://www.vaadata.com/en/blog/understanding-ntlm-authentication-and-ntlm-relay-attacks/>

<https://ponderthebits.com/2018/02/windows-rdp-related-event-logs-identification-tracking-and-investigation/>